



Universität
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An Algorithm to Compute Piecewise Ehrhart Quasi-Polynomials Associated to Matrices

MASTER THESIS

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Abstract

In this thesis, we first look at Ehrhart polynomials of integral polytopes, counting lattice points inside scalings of polytopes. These can be generalised to Ehrhart quasi-polynomials of rational polytopes. We carry our knowledge over to construct piecewise Ehrhart quasi-polynomials associated to systems of linear inequalities with fixed linear part (described by a matrix) and varying constant part. We also give explicit algorithms which were implemented in a software package to go with this thesis¹, which is described at the end [19].

¹The package can be found at <https://github.com/mwooll/EhrhartQuasiPolynomial>

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1 Introduction

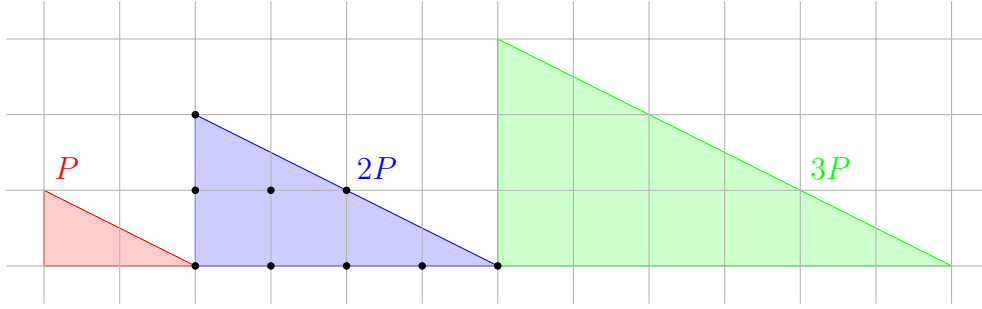


Figure 1: Triangle P and its dilates $2P$ and $3P$

Consider the triangle $P = \text{conv}(\{(0,0), (2,0), (0,1)\})$ and its dilates $2P$ and $3P$ from Figure 1. When we count how many grid points lie in the triangles, we get 4 for P , 9 for $2P$ and 16 for $3P$. Figure 1 shows these explicitly for $2P$.

The number of grid points we count are equal to the squares of 2, 3 and 4, respectively, which means that tP has $(t+1)^2$ grid points for $t = 1, 2, 3$. In fact this relation holds for all $t \in \mathbb{N}$ and we call $(t+1)^2$ the *Ehrhart polynomial* of P denoted $\mathbf{ehr}_P(t)$.

Ehrhart polynomials got their names from Eugène Ehrhart, a French mathematician from Strasbourg, who in the 1960's showed that every integral polytope P has such a polynomial. He even generalized this result by showing that every rational polytope Q has an Ehrhart quasi-polynomial $\mathbf{ehr}_Q(t)$ stating how many integral points tQ has, for precise statements see chapter 3. The theorems of Ehrhart only state the existence of the Ehrhart (quasi-)polynomial so the problem arises how to compute them.

Figure 2 shows the rational triangle Q and its dilates $2Q$ and $3Q$. Its Ehrhart quasi-polynomial is

$$\mathbf{ehr}_Q(t) = \begin{cases} \frac{1}{4}t^2 + t + \frac{3}{4} & \text{for } t \text{ odd} \\ \frac{1}{4}t^2 + t + 1 & \text{for } t \text{ even} \end{cases}$$

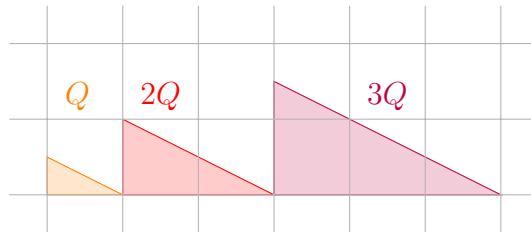


Figure 2: Triangle Q and its dilates $2Q$ and $3Q$

There are optimized implementations to compute the Ehrhart polynomial of integral polytopes, for example in Sage [17] and Normaliz [4]. Normaliz also provides an implementation to compute Ehrhart quasi-polynomials of rational polytopes, which can be used through a Sage interface. The latter option however returns a tuple of polynomials which does not allow to perform meaningful computations in the ring of quasi-polynomials.

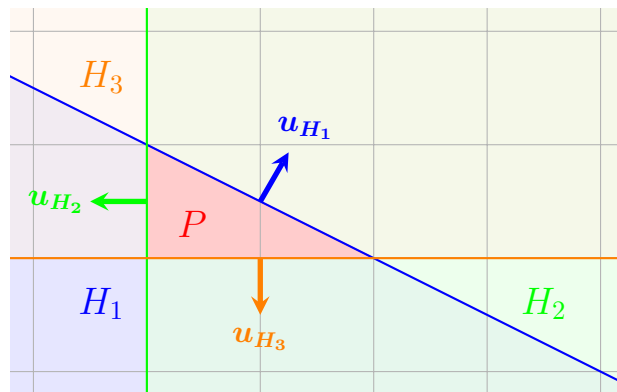


Figure 3: Triangle P and its defining half spaces H_1, H_2 and H_3 with their outward pointing normal vectors.

The first contribution of this thesis is a Sage implementation of quasi-polynomials which allows ring computations over a specifiable commutative base ring, as described in section 2.4.²

The package also provides a basic implementation to compute Ehrhart quasi-polynomials which then return an instance of the mentioned quasi-polynomial class.

A polytope can also be described as the intersection of a set of half-spaces. Figure 3 shows how P is defined through the 3 half-spaces H_1, H_2 and H_3 whose outward pointing normal vectors u_{H_1}, u_{H_2} and u_{H_3} are each perpendicular to one edge of P . With this representation it is natural to ask how the number of grid points changes when we move the lines which define the edges of P independently.

In [10] it was shown that in this case there exists a piecewise multivariate quasi-polynomial which evaluates to the number of grid points of the specific scaling of P . But there are no readily available implementations to compute these.

The second contribution of this thesis is an explicit algorithm to compute piecewise Ehrhart quasi-polynomials associated to matrices, as described in chapter 5. To verify its correctness the algorithm was also implemented in the Sage package.²

To build up to our second contribution we will use the preliminaries sections to introduce most mathematical objects needed for our constructions. Chapter 3 shows the mentioned theorems by Eugène Ehrhart and Pick's Theorem which predates Ehrhart's work by 60 years. In chapter 4 we compute piecewise Ehrhart polynomials in some lower dimensions to get a grasp of the general case. Finally in chapter 5 we derive the algorithm to compute piecewise Ehrhart quasi-polynomials associated to matrices and finish with showcasing the implemented package in chapter 6.²

² The package can be found at <https://github.com/mwooll/EhrhartQuasiPolynomial>

2 Preliminaries

Let $\mathbf{0}$ denote the origin in any appropriate dimension. $\gcd$ and lcm denote the *greatest common divisor* and the *least common multiple*, respectively.

Definition 2.0.1. Given a set $X \subseteq \mathbb{R}^d$, we call sets of the form

$$kX := \{kx \mid x \in X\}, \quad k \in \mathbb{R}$$

dilates of X . And specifically kX the k^{th} dilate of X .

Definition 2.0.2. [1] Given $a, b, k \in \mathbb{R}$, we say a and b are congruent modulo k and write

$$a \equiv b \pmod{k}$$

if $a - b$ is an integer multiple of k . We also may write

$$a \bmod k$$

to mean the unique value $r \in \mathbb{R}$ with

$$a \equiv r \pmod{k} \quad \wedge \quad 0 \leq r < |k|.$$

2.1 Polynomials

Definition 2.1.1. [1] A polynomial function f in one variable over a ring R is a finite linear combination of the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where the coefficients a_i lie in R . The set of all such polynomial functions is denoted $\mathbf{R}[x]$. The degree of a polynomial $\deg(f)$ is the largest number n such that a_n is not 0.

Proposition 2.1.2. [1] $\mathbf{R}[x]$ is a commutative ring, with addition and multiplication of polynomial functions. $\mathbf{R}$ is a subring if we identify its elements with constant polynomial functions. $\square$

Theorem 2.1.3 (Polynomial Interpolation). [16] Let R be a field. Given $n + 1$ data points

$$X := \{(x_i, y_i) \in R^2 \mid x_i \neq x_j \forall i \neq j\}$$

there exists a unique polynomial p over R of degree at most n such that $f(x_i) = y_i \forall i$. $\square$

Definition 2.1.4. Let X and p be as in the above theorem. We call p the interpolating polynomial of X .

There are several ways to compute the interpolating polynomial, for example: Lagrange interpolation and Newton interpolation [16], but since it is unique, we do not care which method is used and choose a suitable one.

For our purposes, we are mostly interested in polynomials over $\mathbb{Q}$, i.e., elements of $\mathbb{Q}[x]$, which often arise as we compute the interpolating polynomials of sets in $\mathbb{Z}^2$.

2.2 Multivariate Polynomials

Since a polynomial ring $\mathbf{R}[x]$ is, as its name suggests, itself a ring, we can use it as a base for new polynomial rings like $\mathbf{R}[x][y]$, $\mathbf{R}[x][y][z]$, etc., we call these *multivariate polynomial rings*. Let $f \in \mathbf{R}[x][y]$, then f is of the form

$$f(x, y) = p_n(x)y^n + \cdots + p_1(x)y + p_0(x)$$

we can rearrange the terms to get the following representation

$$f(x, y) = \sum_{0 \leq i \leq n, 0 \leq j \leq m} a_{i,j} x^i y^j$$

In fact, mathematically there is no difference between $\mathbf{R}[x][y]$ and $\mathbf{R}[y][x]$, for this reason it makes sense to write $\mathbf{R}[x, y]$ or even $\mathbf{R}[x_1, \dots, x_n]$ if we have n variables.

Definition 2.2.1. For $k \in \mathbb{N}$ we denote, as usual, the factorial of k as

$$k! := \prod_{n=1}^k n$$

For $m, k \in \mathbb{N}$ we define the binomial coefficient of m and k as

$$\binom{m}{k} := \frac{m!}{k! \cdot (m-k)!}$$

Theorem 2.2.2. To interpolate a multivariate polynomial of degree at most d in k variables, we need $\binom{d+k}{k}$ points and their evaluations. $\square$

Remark 2.2.3. It is important to note, that not every choice of $\binom{d+k}{k}$ different points will interpolate to the correct multivariate polynomial. Since, for example, if every chosen point lies on the x -axis, then it is impossible to interpolate the influence y has on the polynomial. We will give a suitable choice of points for the cases where we want to perform multivariate interpolation.

2.3 Integer Periodic Function

Definition 2.3.1. [8] An integer periodic function is a function f defined on $\mathbb{Z}$ with values in $\mathbb{Q}$ for which there exists a $p \in \mathbb{N}$ such that

$$f(t+p) = f(t) \quad \forall t \in \mathbb{Z}.$$

We call p the period of f . If $p = 1$ we say that f is a constant integer periodic function. The set of all integer periodic functions is denoted as $\mathbb{Q}_{\mathbb{Z}}$.

We can define an integer periodic function f with period n through the ordered set

$$f = [f(0), f(1), \dots, f(n-1)] = [f(t) \mid 0 \leq t \leq n-1]$$

and evaluate it at arbitrary large integers t as $f(t \bmod n)$.

Definition 2.3.2. [8] Let f and g be two integer periodic functions with periods p and q respectively. We define their sum and product as the integer periodic functions determined by the following ordered sets

$$\begin{aligned} f + g &= [f(t) + g(t) \mid 0 \leq t \leq p \cdot q - 1] \\ f \cdot g &= [f(t) \cdot g(t) \mid 0 \leq t \leq p \cdot q - 1] \end{aligned}$$

Theorem 2.3.3. [8] $\mathbb{Q}_{\mathbb{Z}}$ is a commutative ring with addition and multiplication as given in Definition 2.3.2, for which the constant integer periodic functions 0 and 1 are the neutral elements. $\mathbb{Q}$ is a subring if we identify its elements with constant integer periodic functions. $\square$

Proof. Let f be an integer periodic function with period p .

1) 0 and 1 are the neutral elements since

$$\begin{aligned} f &= [f(0), f(1), \dots, f(n-1)] = [f(0) + 0, f(1) + 0, \dots, f(n-1) + 0] = f + 0 \\ &= [f(0) \cdot 1, f(1) \cdot 1, \dots, f(n-1) \cdot 1] = f \cdot 1 \end{aligned}$$

2) associativity and commutativity of addition follow from associativity and commutativity of addition over $\mathbb{Q}$.

3) the additive inverse of f is determined by

$$-f = [-f(0), -f(1), \dots, -f(n-1)]$$

5) associativity and commutativity of multiplication follow from associativity and commutativity of multiplication over $\mathbb{Q}$.

6) distributivity follows from distributivity over $\mathbb{Q}$. $\square$

Proposition 2.3.4. [8] Let f and g be in $\mathbb{Q}_{\mathbb{Z}}$. The periods of $f + g$ and $f \cdot g$ each divides the least common multiple of the periods of f and g .

Further, if the periods of f and g are coprime

$$\text{per}(f + g) = \text{per}(f) \cdot \text{per}(g)$$

and if additionally neither f nor g is the constant 0

$$\text{per}(f \cdot g) = \text{per}(f) \cdot \text{per}(g). \quad \square$$

Example 2.3.5. Consider the integer periodic functions $f = [0, 1]$ and $g = [1, 0]$, both of period 2.

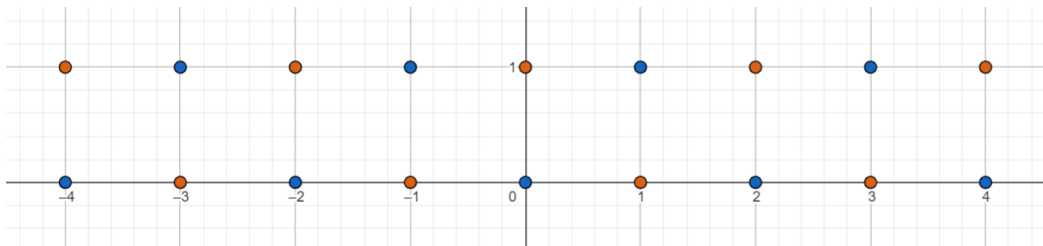


Figure 4: Integer Periodic Functions f (blue) and g (orange)

The sum of f and g is the constant 1 and their product is the constant 0.

2.4 Quasi-Polynomials

Definition 2.4.1. [15] A quasi-polynomial q is a function defined on $\mathbb{Z}$ of the form

$$\begin{aligned} q(t) &= c_d(t)t^d + c_{d-1}(t)t^{d-1} + \cdots + c_0(t) \\ &= [c_{d,0}, \dots, c_{d,n_d-1}]t^d + \cdots + [c_{0,0}, \dots, c_{0,n_0-1}] \end{aligned}$$

where the $c_i = [c_{i,0}, \dots, c_{i,n_i-1}]$ are integer periodic functions in t with periods n_i , called the coefficient functions.

We define the degree of q to be the largest d such that c_d is not identical to 0 and we define its period $\text{per}(q)$ to be the least common multiple of the n_i .

By definition, quasi-polynomials are polynomials over the commutative ring $\mathbb{Q}_{\mathbb{Z}}$. Hence by Proposition 2.1.2 $\mathbb{Q}_{\mathbb{Z}}[t]$ is a commutative ring with addition and multiplication, for which the constants 0 and 1 are the neutral elements. $\mathbb{Q}_{\mathbb{Z}}$ is a subring if we identify its elements with quasi-polynomials of degree 0. $\mathbb{Q}[t]$ is a subring since its coefficients are just constant periodic integer functions.

Proposition 2.4.2. Let f and g be quasi-polynomials. The periods of $f + g$ and $f \cdot g$ each divides the least common multiple of the periods of f and g . $\square$

Remark 2.4.3. [3] Alternatively, a quasi-polynomial q with period k can also be defined through k polynomials $p_0, \dots, p_{k-1}$ as

$$q(t) = \begin{cases} p_0(t) & \text{if } t \equiv 0 \pmod{k} \\ p_1(t) & \text{if } t \equiv 1 \pmod{k} \\ \vdots & \\ p_{k-1}(t) & \text{if } t \equiv k-1 \pmod{k} \end{cases}$$

Given a quasi-polynomial in this form, one can combine coefficients of monomials sharing the same degree into integer periodic functions and get the corresponding representation according to their definition.

We will mostly describe and define quasi-polynomials according to Definition 2.4.1, but Remark 2.4.3 gives us a nice way to define the notion of interpolating quasi-polynomial reminiscent of Theorem 2.1.3:

Construction 2.4.4. Given $p \cdot (n+1)$ data points

$$X := \{(i, y_i) \in \mathbb{N} \times \mathbb{Q} \mid 1 \leq i \leq p \cdot (n+1)\},$$

group the points into sets of $n+1$ points as follows

$$X_k := \{(i, y_i) \mid i \equiv k \pmod{p}\}$$

By Theorem 2.1.3 there exists a unique polynomial p_k of degree at most n passing through the points $X_k \forall 0 \leq k \leq p-1$. These p_k define a quasi-polynomial q as per Remark 2.4.3. We call q the interpolating quasi-polynomial of X .

Since the factorisation of $p(n+1)$ into p and $n+1$ might not be the only factorisation, there may exist interpolating quasi-polynomials with different periods and thus in general also different degrees, for example with period $= n+1$ and degree $\leq p$. But having fixed the size of the X_k , the interpolating quasi-polynomial is unique since the individual interpolating polynomials are unique by Theorem 2.1.3. In the cases where we are interested in the interpolating quasi-polynomial we will determine p beforehand.

2.5 Polytopes

Definition 2.5.1. [5] Let $X \subset \mathbb{R}^d$. A linear combination of X is a sum

$$v := \sum_{x \in X} \lambda_x x$$

where $\lambda_x = 0$ for all but finitely many $x \in X$. The linear span $\text{Span}(X)$ of X is the set of all linear combinations of X .

A linear combination is an affine combination if additionally the sum of the coefficients λ_x is 1. The affine hull $\text{aff}(X)$ of X is the set of all affine combinations.

Definition 2.5.2. [5] A linear combination is conic if all coefficients are non-negative, and it is convex if it is conic and affine.

The set of all convex combinations of a set X is the convex hull $\text{conv}(X)$ of X .

A polytope is the convex hull of finitely many points in $\mathbb{R}^d$. The dimension $\dim(P)$ of a polytope P is the dimension of the affine space it spans.

Note: Polytopes are a generalisation of *polygons*, which are 2 dimensional polytopes.

Definition 2.5.3. [5] For any $\mathbf{0} \neq a \in \mathbb{R}^d$ and $b \in \mathbb{R}$ the set

$$H := \{x \in \mathbb{R}^d \mid \langle a, x \rangle \leq b\}$$

is the affine half space defined by a and b .

Definition 2.5.4. [5] A polyhedron P is the intersection of finitely many affine half spaces

$$P = \{x \in \mathbb{R}^d \mid \langle a_1, x \rangle \leq b_1, \dots, \langle a_n, x \rangle \leq b_n\}$$

for $a_i \in \mathbb{R}^d$ and $b_i \in \mathbb{R} \forall 1 \leq i \leq n$.

Theorem 2.5.5 (Weyl-Minkowski-Duality). [5] A bounded set $P \subset \mathbb{R}^d$ is a polytope if and only if it is the bounded intersection of a finite number of affine half spaces. $\square$

From this theorem we obtain two equivalent descriptions of a polytope [5]:

1. a polytope is the convex hull of a finite set of points in $\mathbb{R}^d$
2. a polytope is the bounded intersection of a finite set of affine half spaces

Both descriptions are useful, and we will choose whichever is more convenient for a given problem.

Remark 2.5.6. Given a polytope P according to Definition 2.5.4:

$$P = \{x \in \mathbb{R}^d \mid \langle a_1, x \rangle \leq b_1, \dots, \langle a_n, x \rangle \leq b_n\}$$

we condense this to

$$P = P_A(b) := \{x \mid Ax \leq b\}$$

where the inequality is understood componentwise and

$$A = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \text{Mat}(n \times d, \mathbb{R}), \quad b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in \mathbb{R}^n.$$

Now that we have established what polytopes are, we will need to define some further properties to make them useful to us.

Definition 2.5.7. *Given a polytope P , we denote its interior by $\mathring{P}$ and its boundary by ∂P , which fulfil the expected equalities:*

$$\begin{aligned}\mathring{P} \cup \partial P &= P \\ \mathring{P} \cap \partial P &= \emptyset\end{aligned}$$

Definition 2.5.8. [5] *Given a polytope $P \subset \mathbb{R}^d$, we call $x \in P$ an integral point of P if $x \in \mathbb{Z}^d$. The set of all integral points of P is $P \cap \mathbb{Z}^d$. And we call x an interior point if $x \in \mathring{P}$.*

Definition 2.5.9. [5] *A polytope P in $\mathbb{R}^d$ is called rational if all its vertices lie in $\mathbb{Q}^d$ and it is called integral if all its vertices lie in $\mathbb{Z}^d$.*

Definition 2.5.10. *For an integral polytope P defined by the vertices $X \subset \mathbb{Z}^d$, we define*

$$\gcd(P) := \gcd\{x_{i,j} \mid x_i \in X, 1 \leq j \leq d\}$$

That is $\frac{1}{\gcd(P)}P$ is the smallest dilate of P which is integral.

Definition 2.5.11. *For a rational polytope P we define its period $\text{per}(P)$ to be the smallest natural number $0 \neq k \in \mathbb{N}$ such that kP is an integral polytope.*

Remark 2.5.12. *The period of a polytope can be computed efficiently as the least common multiple of the denominators of the coordinates of all its vertices.*

Example 2.5.13. *Let P be the polytope defined by the vertices*

$$\left\{ \left(-\frac{1}{4}, 0 \right), \left(\frac{2}{3}, 0 \right), \left(0, \frac{1}{10} \right) \right\}$$

Its period is $\text{lcm}(4, 3, 10) = 4 \cdot 3 \cdot 5 = 60$. Therefore, kP is integral $\iff k$ is divisible by 60. The only integral point of P is $(0, 0)$.

When given the vertices of a polytope directly, the period can be computed at a glance, however given a polytope according to Remark 2.5.6 this is not immediate.

Example 2.5.14. *Let P be the polytope defined by the following set of linear inequalities*

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Since only integers are present in the linear inequalities, one might expect this polytope to be integral. However, it has the vertices

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix} \right\}$$

and hence a period of 2.

Definition 2.5.15. [14] A polytope Q is called a face of P , if either $Q = P$ or we can write

$$Q = P \cap \partial H$$

with a closed half space $P \subset H$. In that case we write $Q \preceq P$. Faces of dimensions 0, 1 and $\dim(P) - 1$ are called vertices, edges and facets, respectively.

Given the polytope $P := P_A(b)$, we can obtain any facet F of a polytope P as an intersection of the form

$$P \cap \{x \in \mathbb{R}^d \mid \langle a_F, x \rangle = b_F\},$$

where a_F is a row vector of A and b_F the corresponding value of b . In this case a_F is the outward pointing normal vector of F . The inward pointing normal vector $u_F = -a_F$ can also be used to define P and F as

$$P = \{x \mid \langle u_F, x \rangle \geq -b_F\}, \quad F = P \cap \{x \mid \langle u_F, x \rangle = b_F\}.$$

2.6 Bounding Box

Definition 2.6.1. For a polytope P of dimension d defined by the set of vertices V , we define its bounding box to be the polytope

$$\{(b_1, \dots, b_d) \in \mathbb{R}^d \mid b_j \in [\min_j(P), \max_j(P)] \forall 1 \leq j \leq d\}$$

where

$$\begin{aligned} \min_j(P) &:= \min(\{v_{i,j} \mid v_i \in V\}), \\ \max_j(P) &:= \max(\{v_{i,j} \mid v_i \in V\}) \end{aligned}$$

are the minimal and maximal values present across the j^{th} coordinate of the vertices in V . The integral points of the bounding box of P are given by the set

$$\{(b_1, \dots, b_d) \in \mathbb{Z}^d \mid \min_j(P) \leq b_j \leq \max_j(P) \forall 1 \leq j \leq d\}. \quad (1)$$

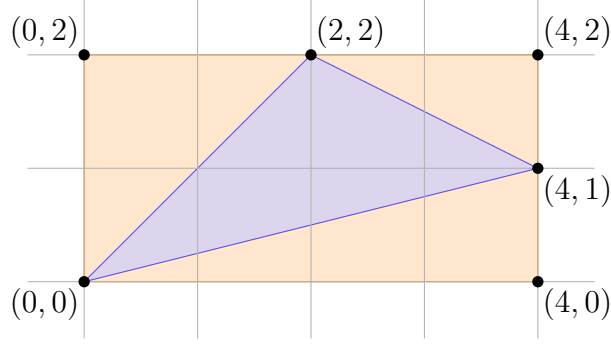
Note: by design, P is a subset of the bounding box of P , therefore, all integral points of P lie inside its bounding box.

Example 2.6.2. Let T be the triangle given by the vertices

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\}$$

then its bounding box is the rectangle with these vertices:

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$$


 Figure 5: Bounding box of the Triangle T

2.7 Counting Points on Line Segments

Theorem 2.7.1. *Let L be the line segment from $A = \begin{pmatrix} x_A \\ y_A \end{pmatrix} \in \mathbb{Z}^2$ to $B = \begin{pmatrix} x_B \\ y_B \end{pmatrix} \in \mathbb{Z}^2$. Then*

$$|L \cap \mathbb{Z}^2| = \gcd(|x_B - x_A|, |y_B - y_A|) + 1$$

Proof. Let L' be the line segment from $\mathbf{0}$ to $B' = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix} \in \mathbb{Z}^2$ and let M be the line segment from $\mathbf{0}$ to $C = \begin{pmatrix} |x_B - x_A| \\ |y_B - y_A| \end{pmatrix} =: \begin{pmatrix} x_C \\ y_C \end{pmatrix} \in \mathbb{N}_0^2$. There exist the bijections

$$\begin{aligned} \varphi : L &\longleftrightarrow L' \\ p &\longleftrightarrow p - A \end{aligned}$$

and

$$\begin{aligned} \phi : L' &\longleftrightarrow M \\ \begin{pmatrix} x \\ y \end{pmatrix} &\longleftrightarrow \begin{pmatrix} |x| \\ |y| \end{pmatrix} \end{aligned}$$

which tell us

$$|L \cap \mathbb{Z}^2| \stackrel{\varphi}{=} |L' \cap \mathbb{Z}^2| \stackrel{\phi}{=} |M \cap \mathbb{Z}^2|.$$

Assume $x_C = 0$, then M is isomorphic to the interval $[0, y_C]$ and thus

$$|M \cap \mathbb{Z}^2| = y_C + 1.$$

Let now $x_C \neq 0$. Note that for $p = \begin{pmatrix} x_p \\ y_p \end{pmatrix} \in \mathbb{N}_0^2$ we have

$$p \in M \iff x_p \leq x_C \wedge y_p \leq x_C \wedge \frac{y_p}{x_p} = \frac{y_C}{x_C}.$$

Let $d := \gcd(x_C, y_C)$ and $x_d := \frac{x_C}{d}, y_d := \frac{y_C}{d}$ hence

$$\frac{y_C}{x_C} = \frac{d \cdot y_d}{d \cdot x_d} = \frac{y_d}{x_d}.$$

Therefore all $p \in M \cap \mathbb{Z}^2$ can be written as $k \cdot \frac{y_d}{x_d}$ for some $0 \leq k \leq d$. Hence

$$\begin{aligned} |M \cap \mathbb{Z}^2| &= \left| \left\{ k \cdot \frac{y_d}{x_d} \mid k \in \mathbb{N}_0 \wedge 0 \leq k \leq d \right\} \right| \\ &= d + 1 \\ &= \gcd(x_C, y_C) + 1 \\ &= \gcd(|x_B - x_A|, |y_B - y_A|) + 1. \end{aligned}$$

Since $\gcd(n, 0) = n \forall n \in \mathbb{N}_0$ we get

$$|L \cap \mathbb{Z}^2| = \gcd(|x_B - x_A|, |y_B - y_A|) + 1.$$

□

Theorem 2.7.2. *Let L be the line segment from $A = \begin{pmatrix} x_A \\ y_A \end{pmatrix} \in \mathbb{Z}^2$ to $B = \begin{pmatrix} x_B \\ y_B \end{pmatrix} \in \mathbb{Z}^2$ and tL the line segment from tA to tB . Then*

$$|tL \cap \mathbb{Z}^d| = |t|(|L \cap \mathbb{Z}^2| - 1) + 1 \quad \forall t \in \mathbb{Z}.$$

Proof. We know from Theorem 2.7.1,

$$|L \cap \mathbb{Z}^2| = \gcd(|x_B - x_A|, |y_B - y_A|) + 1.$$

And therefore

$$\begin{aligned} |tL \cap \mathbb{Z}^2| &= \gcd(|t \cdot x_B - t \cdot x_A|, |t \cdot y_B - t \cdot y_A|) + 1 \\ &= \gcd(|t| \cdot |x_B - x_A|, |t| \cdot |y_B - y_A|) + 1 \\ &= |t| \cdot \gcd(|x_B - x_A|, |y_B - y_A|) + 1 \\ &= |t| \cdot (\gcd(|x_B - x_A|, |y_B - y_A|) + 1 - 1) + 1 \\ &= |t| \cdot (|L \cap \mathbb{Z}^2| - 1) + 1. \end{aligned}$$

□

2.8 Lattice

Definition 2.8.1. [2] *A subset $\Lambda \subset \mathbb{R}^n$ is called a lattice, if*

- i Λ is an additive subgroup of $\mathbb{R}^n$: for all $x, y \in \Lambda$ we have $x \pm y \in \Lambda$;*
- ii Λ is discrete, that is, for every bounded set $B \subset \mathbb{R}^n$, the intersection $B \cap \Lambda$ is finite;*
- iii Λ spans $\mathbb{R}^n$: $\text{Span}(\Lambda) = \mathbb{R}^n$.*

If $\Lambda \subset \mathbb{Z}^n$ holds, we call Λ an integral lattice.

To satisfy condition *iii*, we need that $\Lambda \subset \mathbb{Z}^n$ contains n linearly independent vectors $v_i \in \mathbb{Z}^n$. So given Λ we can find n such vectors $(v_i)_{1 \leq i \leq n} \subset \mathbb{Z}^n$ and call them a *basis* of Λ . Reversely, any n independent vectors u_i in $\mathbb{Z}^n$ form the basis of some integral lattice Λ . This also means that $\mathbb{Z}^n$ itself is a lattice for all n .

Using the existence of a basis of any lattice, we define the following:

Definition 2.8.2. [2] *Let Λ be a lattice and $u_1, \dots, u_n$ a basis of Λ . The semi-open parallelepiped*

$$\Pi = \Pi(u_1, \dots, u_n) = \left\{ \sum_{i=1}^n \alpha_i u_i \mid 0 \leq \alpha_i < 1 \forall 1 \leq i \leq n \right\}$$

is called a fundamental parallelepiped of Λ .

The fundamental parallelepiped has some useful properties:

Lemma 2.8.3. [2] *Let Λ be a lattice and Π a fundamental parallelepiped of Λ . Then every point $x \in \mathbb{R}^n$ can be uniquely written as*

$$x = y + u$$

where $y \in \Pi$ and $u \in \Lambda$. □

Given $x = y + u$ as in the lemma, we call y the *representative* of x and write

$$\bar{x} := y.$$

Theorem 2.8.4. [2] *Let Λ be a lattice. Then all the fundamental parallelepiped Π of Λ have the same volume, called the determinant of Λ and denoted $\det(\Lambda)$.* □

Let Λ_0, Λ be lattices such that $\Lambda_0 \subset \Lambda$. Then Λ_0 is a subgroup of Λ and we can consider the quotient Λ/Λ_0 and cosets of Λ modulo Λ_0 [2].

Theorem 2.8.5. [2] *Let $\Lambda_0 \subset \Lambda$ be lattices and let Π_0 be a fundamental parallelepiped of Λ_0 . Then the set $\Pi_0 \cap \Lambda$ contains each coset Λ/Λ_0 representative exactly once. Furthermore*

$$|\Pi_0 \cap \Lambda| = |\Lambda/\Lambda_0| = \frac{\det(\Lambda_0)}{\det(\Lambda)}. \quad \square$$

This means we have a bijection between $\mathbb{Z}^n/\Lambda$ and $\Pi \cap \mathbb{Z}^n$ which we will make use of.

Remark 2.8.6. *Since fundamental parallelepipeds are bounded and lattices are discrete (condition *ii* of Definition 2.8.1), we can conclude that Λ/Λ_0 is finite for all lattices $\Lambda_0 \subset \Lambda$. In particular, $\mathbb{Z}^n/\Lambda$ is finite for all integral lattices Λ .*

2.9 Multivariate Quasi-Polynomials

As we generalized polynomials to multivariate polynomials we also want to generalize quasi-polynomials to *multivariate quasi-polynomials*.

Since we defined quasi-polynomials to be elements of the polynomial ring $\mathbb{Q}_{\mathbb{Z}}[x]$, one way to define multivariate quasi-polynomials is to extend the ring $\mathbb{Q}_{\mathbb{Z}}$ to have multiple independent periods and use this as the basis for the multivariate quasi-polynomial ring.

Definition 2.9.1. A multi periodic function with n periods is a function f defined on $\mathbb{Z}^n$ with values in $\mathbb{Q}$ for which there exists an integral lattice $\Lambda \subset \mathbb{Z}^n$ such that

$$f(\mathbf{t} + u) = f(\mathbf{t}) \quad \forall u \in \Lambda \forall \mathbf{t} \in \mathbb{Z}^n.$$

We denote by $\mathbb{Q}_{\mathbb{Z}^n}$ all multi periodic functions with n periods.

Proposition 2.9.2. $\mathbb{Q}_{\mathbb{Z}^n}$ is a commutative ring with addition and multiplication for which the constant integer periodic functions 0 and 1 are the neutral elements $\forall n \in \mathbb{N}$. $\mathbb{Q}_{\mathbb{Z}}$ is a subring of $\mathbb{Q}_{\mathbb{Z}^n} \forall n \in \mathbb{N}$. □

We call elements of the polynomial ring $\mathbb{Q}_{\mathbb{Z}^n}[\mathbf{x}]$ *multivariate quasi-polynomials*.

One way to define a multivariate quasi-polynomial q which is periodic on Λ , is to define it through a set of polynomials, each with one corresponding point $x_i \in \Pi \cap \mathbb{Z}^n \simeq \mathbb{Z}^n / \Lambda$, where Π is any fundamental parallelepiped of Λ , as

$$q(t) := \begin{cases} p_1(t) & \text{if } \bar{t} = x_1 \\ p_2(t) & \text{if } \bar{t} = x_2 \\ \vdots & \\ p_L(t) & \text{if } \bar{t} = x_L \end{cases} \quad (2)$$

3 Ehrhart Theory

In this chapter we lay out the theory of Ehrhart quasi-polynomials as developed by Eugène Ehrhart and the special case proven by Georg Alexander Pick 60 years prior. For examples of Ehrhart quasi-polynomials see section 3.5 and chapter 4.

3.1 Ehrhart Polynomial

3.1.1 Theoretical Foundation

Definition 3.1.1. [5] *The Ehrhart counting function of a bounded subset $S \subset \mathbb{R}^d$ is the function*

$$\begin{aligned} \mathbf{ehr}_S(k) : \mathbb{Z}_{\geq 1} &\longrightarrow \mathbb{Z}_{\geq 0} \\ k &\longmapsto |kS \cap \mathbb{Z}^d| \end{aligned}$$

Given a polytope P we define its volume $\text{vol}(P)$ as its Lebesgue measure. For a construction of the Lebesgue measure see [13].

Theorem 3.1.2 (Ehrhart's Theorem). [5] *If P is an integral polytope of dimension d , the Ehrhart counting function given by $k \mapsto \mathbf{ehr}_P(k)$ for $k \in \mathbb{N}$ extends to a polynomial function $t \mapsto \mathbf{ehr}_P(t)$ of degree d and leading coefficient $\text{vol}(P)$. $\square$*

For an integral polytope P we call the polynomial $\mathbf{ehr}_P(t)$ its *Ehrhart polynomial*.

Theorem 3.1.3. [5] *Let P be a d -dimensional integral polytope. Then the constant term of the Ehrhart polynomial is 1 if P is non-empty. $\square$*

3.1.2 Algorithm

Abstractly speaking, calculating the Ehrhart polynomial of a given integral polytope P is quite straightforward: count the number of points in $\dim(P) + 1$ different dilates of P and compute from those quantities the unique interpolation polynomial. This is the Ehrhart polynomial of P .

```
EhrhartPolynomial( $P$ ):
  for  $1 \leq k \leq \dim(P) + 1$ :
     $\text{count}_k \leftarrow \text{countIntegralPoints}(kP)$ 
   $X \leftarrow \{(k, \text{count}_k) \in \mathbb{Z}^2 \mid 1 \leq k \leq \dim(P) + 1\}$ 
  return interpolatePolynomial( $X$ )
```

For $\dim(P)$ we use the dimension of the ambient space of P . Calculating the true dimension of P according to Definition 2.5.2 would be quite expensive and often not worth it, since if P is fully dimensional the computations would be unnecessary.

When counting integral points of kP , we need to check at most the points inside the bounding box of kP . To check if a point x lies inside kP , one uses the $\mathcal{H}$ -description of

kP and checks if x lies in every half space of kP .

Note: this is a basic approach to count the integral points, one could also decompose the polytope into simplices and use one of the algorithms described here [7].

```
countIntegralPoints( $kP$ ):
   $count \leftarrow 0$ 
  for point in BoundingBoxPoints( $kP$ ):
    if point in  $kP$ :
       $count \leftarrow count + 1$ 
  return  $count$ 
```

BoundingBoxPoints(kP) generates all the integral points of the bounding box of kP , i.e. the set of points described in formula (1).

Lastly, we need to compute the unique interpolation polynomial of the points

$$\{(k, count_k) \in \mathbb{Z}^2 \mid 1 \leq k \leq \dim(P) + 1\}$$

We know from Theorem 2.1.3 that given $n + 1$ points such that $x_i \neq x_j \forall i \neq j$, the interpolation polynomial will be of degree at most n . Thus $\dim(P) + 1$ points will suffice since the polynomial could have a degree of at most $\dim(P)$ according to Theorem 3.1.2.

3.2 Ehrhart Quasi-Polynomial

3.2.1 Theoretical Foundation

There exists a nice extension of Theorem 3.1.2 for rational polytopes.

Theorem 3.2.1 (Ehrhart's Theorem for Rational Polytopes). [3] *If P is a rational polytope of dimension d , then $\mathbf{ehr}_P(t)$ is a quasi-polynomial of degree d and its period divides the period of P .* $\square$

For a rational polytope P , we call $\mathbf{ehr}_P(t)$ its *Ehrhart quasi-polynomial*.

3.2.2 Algorithm

Since now we want to compute Ehrhart quasi-polynomial for rational polytopes, we need to make some changes to the algorithm described in section 3.1.2.

Let P be a rational polytope with period p for which we want to compute the Ehrhart quasi-polynomial q . We know from Construction 2.4.4 that we can compute a quasi-polynomial with period p and degree d with exactly $p \cdot (d + 1)$ points and since Theorem 3.2.1 tells us that $\deg(q) \leq d$ and $\text{per}(q) \mid p$, in particular $\text{per}(q) \leq p$. Therefore, if we count the integral points of exactly $p \cdot (d + 1)$ dilates of P and interpolate p polynomials which we combine into a quasi-polynomial q of degree $\leq d$, we have computed the Ehrhart quasi-polynomial of P .

```
EhrhartQuasiPolynomial( $P$ ):
  for  $1 \leq k \leq \text{per}(P) \cdot (\dim(P) + 1)$ :
```

```

    countk ← countIntegralPoints(kP)
    X ← {(k, countk) ∈ ℤ2 | 1 ≤ k ≤ per(P) · (dim(P) + 1)}
    return interpolateQuasiPolynomial(X, P)

```

We compute $\text{per}(P)$ as noted in Remark 2.5.12. No changes are needed for $\text{dim}(P)$, $\text{countIntegralPoints}$ and BoundingBoxPoints . However since in general we expect a quasi-polynomial, we need a new interpolation algorithm.

```

interpolateQuasiPolynomial(X, P):
  for 1 ≤ k ≤ per(P):
    Xk := {(i, counti) ∈ X | i ≡ k mod per(P)}
    pk ← interpolatePolynomial(Xk)
  return constructQuasiPolynomial({pk | 1 ≤ k ≤ per(P)})

```

The interpolation polynomials p_k define a quasi-polynomial according to Remark 2.4.3, $\text{constructQuasiPolynomial}$ merges coefficients of matching degrees into integer-periodic functions and returns the quasi-polynomial according to Definition 2.4.1.

3.3 Pick's Theorem

Now that we have seen the theorems by Eugène Ehrhart, we should also take a look at Pick's Theorem which predates the shown work of Ehrhart by about 60 years.

Theorem 3.3.1 (Pick's Theorem). [3] *Given an integral convex polygon $P \in \mathbb{R}^2$, and denoting the number of interior integral points by I_P and the number of integral points on the boundary by B_P , we can compute its area A_P as:*

$$A_P = I_P + \frac{1}{2}B_P - 1.$$

□

We can rearrange this formula to compute the number of interior integral points of P :

$$I_P = A_P - \frac{1}{2}B_P + 1$$

or if we wanted the number of all integral points of P :

$$I_P + B_P = A_P + \frac{1}{2}B_P + 1.$$

For $t \in \mathbb{N}$, tP is still an integral convex polygon and hence

$$I_{tP} + B_{tP} = A_{tP} + \frac{1}{2}B_{tP} + 1,$$

since we just scaled every point by t . This also means that the edge lengths are scaled by t but the area is scaled by t^2 , i.e.:

$$A_{tP} = A_P t^2.$$

For B_{tP} we can look at every edge $E_{tP,i} \in B_{tP}$ individually and use Theorem 2.7.2 to get

$$|E_{tP,i} \cap \mathbb{Z}^2| = t(|E_{P,i} \cap \mathbb{Z}^2| - 1) + 1 \quad \forall E_{tP,i} \in B_{tP}.$$

If we just add these up, we would count every vertex twice, so we need to subtract the number of vertices from the sum. And since we are dealing with a 2 dimensional polytope, the number of vertices is equal to the number of edges which does not change when we scale the polytope.

$$\begin{aligned}
B_{tP} &= \sum_{E_{tP,i} \in B_{tP}} |E_{tP,i} \cap \mathbb{Z}^2| - \sum_{E_{P,i} \in B_P} 1 \\
&= \sum_{E_{P,i} \in B_P} t(|E_{P,i} \cap \mathbb{Z}^2| - 1) + 1 - \sum_{E_{P,i} \in B_P} 1 \\
&= t \sum_{E_{P,i} \in B_P} |E_{P,i} \cap \mathbb{Z}^2| - 1 + \sum_{E_{P,i} \in B_P} 1 - \sum_{E_{P,i} \in B_P} 1 \\
&= t \sum_{E_{P,i} \in B_P} |E_{P,i} \cap \mathbb{Z}^2| - 1 \\
&= B_P t
\end{aligned}$$

So we get

$$I_{tP} + B_{tP} = A_P t^2 + \frac{1}{2} B_P t + 1$$

which is the general formula for the Ehrhart polynomial of P [3]:

$$\mathbf{ehr}_P(t) = A_P t^2 + \frac{1}{2} B_P t + 1. \quad (3)$$

See section 4.3 for an example.

3.4 Equivalence of Ehrhart Quasi-Polynomials

Theorem 3.4.1. *Let P in $\mathbb{R}^d$ be a polytope and $m \in \mathbb{N}$. Then*

$$\mathbf{ehr}_{mP}(t) = \mathbf{ehr}_P(m \cdot t) \quad \forall t \in \mathbb{N}.$$

Proof. Consider the sequences $X := (x_t)_{t \in \mathbb{N}}$ and $Y := (y_t)_{t \in \mathbb{N}}$ where

$$x_t = |t \cdot mP \cap \mathbb{Z}^d|, \quad y_t = |tP \cap \mathbb{Z}^d|$$

By construction, $\mathbf{ehr}_{mP}(t)$ and $\mathbf{ehr}_P(t)$ are the interpolation quasi-polynomials of X and Y respectively. Therefore

$$\mathbf{ehr}_{mP}(t) = |t \cdot mP \cap \mathbb{Z}^d| = |m \cdot tP \cap \mathbb{Z}^d| = \mathbf{ehr}_P(m \cdot t) \quad \forall t \in \mathbb{N}.$$

□

3.5 Example

Let us consider the polytope P defined by

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

which has the following vertices

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \frac{1}{2} \end{pmatrix} \right\}.$$

Since P is embedded in $\mathbb{R}^2$ Theorem 3.1 $\mathbf{ehr}_P(t)$ is of degree at most 2 and specifically the quadratic coefficient is $\frac{1}{4}$ since that is the area of P .

As stated in Example 2.5.14, P has a period of 2, hence the period of $\mathbf{ehr}_P(t)$ needs to divide 2 by Theorem 3.2.1. Therefore we need to interpolate 2 polynomials of degree 2 and thus need to count the integral points of 6 different dilates of P . Where one polynomial describes the behaviour for odd t and the other for even t .

t	1	2	3	4	5	6
$ tP \cap \mathbb{Z}^2 $	2	4	6	9	12	16

Table 1: Number of integral points of tP

So we fit one polynomial through the points

$$X_{odd} = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \begin{pmatrix} 5 \\ 12 \end{pmatrix} \right\}$$

and the other through

$$X_{even} = \left\{ \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 4 \\ 9 \end{pmatrix}, \begin{pmatrix} 6 \\ 16 \end{pmatrix} \right\}$$

Fitting a polynomial each through X_{odd} and X_{even} yields:

$$\begin{aligned} p_{odd} &= \frac{1}{4}t^2 + t + \frac{3}{4} \\ p_{even} &= \frac{1}{4}t^2 + t + 1 \end{aligned}$$

So the two fitted polynomials are almost the same, only having different constant terms. But crucially the quadratic term of both is $\frac{1}{4}t^2$ as we have deduced above. Piecing the two polynomials together into one quasi-polynomial we receive

$$\mathbf{ehr}_P(t) = \begin{cases} \frac{1}{4}t^2 + t + \frac{3}{4} & \text{for } t \text{ odd} \\ \frac{1}{4}t^2 + t + 1 & \text{for } t \text{ even} \end{cases}$$

which we will rewrite to

$$\mathbf{ehr}_P(t) = \frac{1}{4}t^2 + t + \left[1, \frac{3}{4}\right].$$

By Theorem 3.4.1 the Ehrhart quasi-polynomial of $2P$ is just

$$\mathbf{ehr}_{2P}(t) = \mathbf{ehr}_P(2t) = \frac{1}{4}t^2 + t + 1$$

which is just a polynomial.

4 Ehrhart Polynomials of some Families of Polytopes

In this chapter we study the Ehrhart polynomials of polytopes defined by $Ax \leq b$ by varying some components of the right-hand side vector b . Since this will introduce new variables, we will make the convention that t is always used to denote the scaling parameter of the Ehrhart polynomial.

4.1 Right Isosceles Triangle

To reduce our problem to only 1 additional unknown, we consider for $b \in \mathbb{Z}$ the polytope T_b defined by

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \\ b \end{pmatrix}$$

For $b < 0$, the polytope is empty, so $\mathbf{ehr}_{T_b}(t) = 0 \forall b < 0$.

For $b = 0$, we get the origin, so $\mathbf{ehr}_{T_0}(t) = 1$.

For $b > 0$, we get the triangle with the vertices

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} b \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ b \end{pmatrix} \right\}$$

which is a right isosceles triangle. Calculating the Ehrhart polynomial for $b = 1$ yields

$$\mathbf{ehr}_{T_1}(t) = \frac{1}{2}t^2 + \frac{3}{2}t + 1$$

which is already enough to know $\mathbf{ehr}_{T_b}(t)$, since by Theorem 3.4.1 we know

$$\mathbf{ehr}_{T_b}(t) = \mathbf{ehr}_{T_1}(bt).$$

Hence

$$\mathbf{ehr}_{T_b}(t) = \frac{1}{2}b^2t^2 + \frac{3}{2}bt + 1.$$

4.2 Quadrilateral

To get a more interesting example, we consider the polytope $Q_{m,n}$ which satisfies the following set of linear inequations:

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \\ m \\ n \end{pmatrix}$$

This looks quite similar to how we defined T_b , to be exact we added the inequality $y \leq n$. Which is less restricting than $x + y \leq m$ if we choose n larger than m . Hence for $m \leq n$, all points that satisfy the first three inequalities also satisfy the last one. From this we conclude

$$\mathbf{ehr}_{Q_{m,n}}(t) = \mathbf{ehr}_{T_m}(t) \quad \forall m \leq n$$

Now let $m > n$: if we set $n = 0$, y can only be 0, thus

$$x + y \leq m \iff x \leq m$$

this means that our polytope is just the interval $[0, m]$ which has $m + 1$ integral points and an Ehrhart polynomial of $\mathbf{ehr}_{Q_{m,0}}(t) = 1 + mt$.

For $m > n > 0$, $Q_{m,n}$ is a quadrilateral with the following geometry

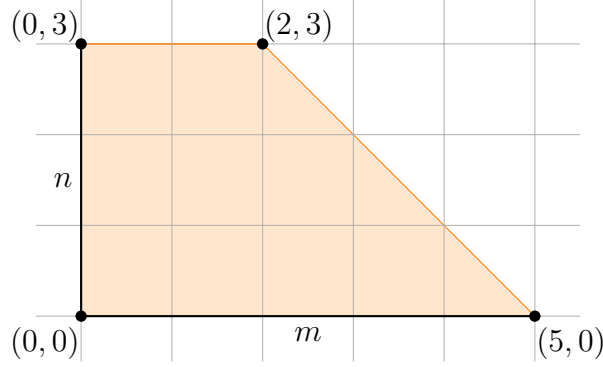


Figure 6: Plot showing $Q_{5,3}$

Note that our quadrilateral is embedded in 2 dimensions, therefore its Ehrhart polynomial has at most degree 2. Further due to Theorem 3.1.2, its leading coefficient is its area which is

$$A_{Q_{m,n}} = mn - \frac{1}{2}n^2$$

and its constant term is 1, since it is non-empty.

Since $Q_{m,n}$ is an integral convex polytope, Pick's theorem 3.3.1 applies, and so does equation (3), which gives us

$$\mathbf{ehr}_{Q_{m,n}}(t) = 1 + \frac{1}{2}B_{Q_{m,n}}t + A_{Q_{m,n}}t^2.$$

Now, $B_{Q_{m,n}}$ is composed of 4 edges, namely:

$$a : \begin{pmatrix} 0 \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} m \\ 0 \end{pmatrix}, b : \begin{pmatrix} m \\ 0 \end{pmatrix} \rightarrow \begin{pmatrix} m-n \\ n \end{pmatrix}, c : \begin{pmatrix} m-n \\ n \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ n \end{pmatrix}, d : \begin{pmatrix} 0 \\ n \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Where a has $m + 1$, b has $n + 1$, c has $m - n + 1$ and d has $n + 1$ integral points. If we add these up and remove 4 due to double counting the vertices we get:

$$\begin{aligned} B_{Q_{m,n}} &= |a \cap \mathbb{Z}^2| + |b \cap \mathbb{Z}^2| + |c \cap \mathbb{Z}^2| + |d \cap \mathbb{Z}^2| - 4 \\ &= (m + 1) + (n + 1) + (m - n + 1) + (n + 1) - 4 \\ &= 2m + n. \end{aligned}$$

So, for $0 < n < m$

$$\begin{aligned} \mathbf{ehr}_{Q_{m,n}}(t) &= 1 + \frac{1}{2}B_{Q_{m,n}}t + A_{Q_{m,n}}t^2 \\ &= 1 + \left(m + \frac{1}{2}n\right)t + \left(mn - \frac{1}{2}n^2\right)t^2. \end{aligned}$$

In conclusion, we know for any combination of $m, n \in \mathbb{Z}$ the Ehrhart polynomial $\mathbf{ehr}_{Q_{m,n}}(t)$:

conditions on m and n	$\mathbf{ehr}_{Q_{m,n}}(t)$
$m < 0$ or $n < 0$	0
$m = 0, \quad n \geq 0$	1
$m \geq 0, \quad n = 0$	$1 + mt$
$0 \leq m \leq n$	$1 + \frac{3}{2}mt + \frac{1}{2}m^2t^2$
$0 \leq n \leq m$	$1 + \left(m + \frac{1}{2}n\right)t + \left(mn - \frac{1}{2}n^2\right)t^2$

Table 2: Explicit formulas for $\mathbf{ehr}_{Q_{m,n}}(t)$

Note that giving a formula for $\mathbf{ehr}_{Q_{m,n}}(t)$ for $m = 0, n \geq 0$ is redundant, since plugging in $m = 0$ into any other valid polynomial also gives 1. Similarly, the case $m \geq 0, n = 0$ is covered by $0 \leq n \leq m$.

So, when we remove these redundancies $\mathbf{ehr}_{Q_{m,n}}(t)$ becomes a rather simple function parametrized by m and n .

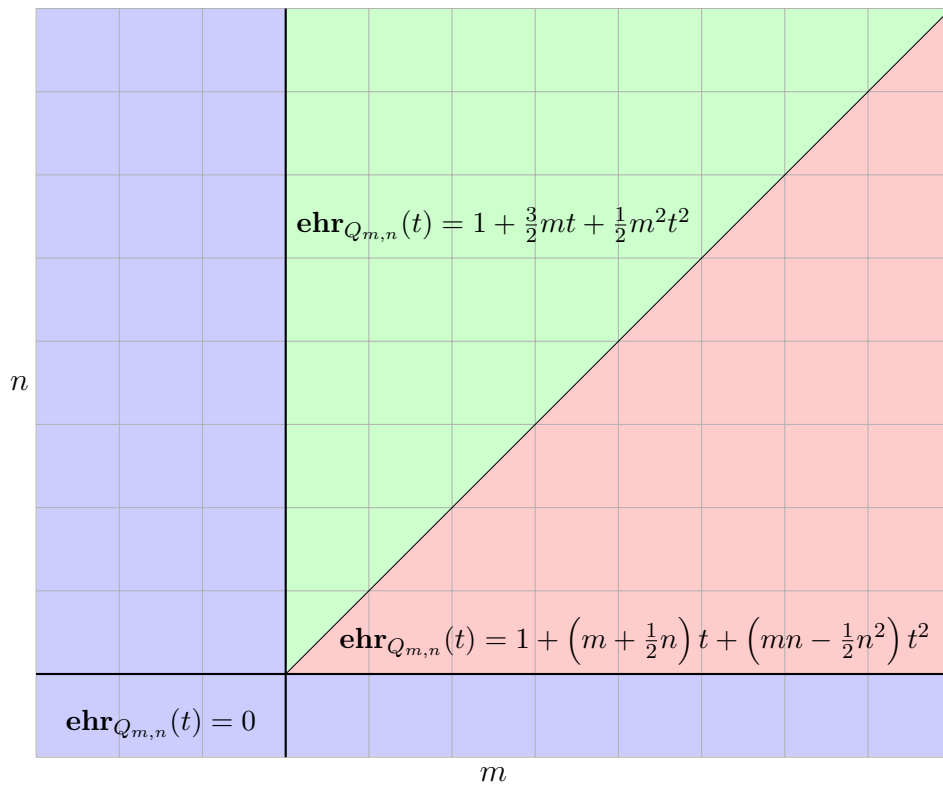


Figure 7: Regions of $\mathbf{ehr}_{Q_{m,n}}(t)$ as a function of m and n

4.3 Right Triangle

Returning to triangles, we can add one unknown by considering the triangle T_{mn} with the vertices

$$\left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} m \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ n \end{pmatrix} \right\}.$$

where m and n lie in $\mathbb{N}$. We will denote the vertices by O, M and N respectively. T_{mn} satisfies the following set of linear inequalities

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \\ \frac{1}{m} & \frac{1}{n} \end{pmatrix} \cdot x \leq \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Note: starting with T_{mn} , we can get T_{nm} by swapping the two coordinates of the vertices, therefore we know that

$$\mathbf{ehr}_{T_{mn}}(t) = \mathbf{ehr}_{T_{nm}}(t)$$

from this we conclude that $\mathbf{ehr}_{T_{mn}}(t)$ must be symmetric with respect to m and n . From Theorem 3.1.2 we know that the leading coefficient is equal to the volume of T_{mn} , since we are in 2 dimensions this is the area, which is $\frac{mn}{2}$. And Corollary 3.1.3 tells us that the constant term is 1 since T_{mn} is not empty. Combining this we get that

$$\mathbf{ehr}_{T_{mn}}(t) = 1 + q(m, n)t + \frac{mn}{2}t^2$$

where $q(m, n)$ is a symmetric function in m and n .

However, this is where we get stuck with our previous approach. Since the parameters m and n are present in the matrix and not in the right-hand side vector, we actually don't expect $q(m, n)$ to be a polynomial.

But thankfully we have also seen Pick's Theorem 3.3.1, which tells us that $q(m, n)$ is equal to half of the number of integral points on ∂T_{mn} , the boundary of T_{mn} , since $T_{m,n}$ is a convex polytope. ∂T_{mn} is composed of the 3 line segments $\overline{OM}$, $\overline{ON}$ and $\overline{MN}$. Since $\overline{OM}$ is part of an axis we can compute the number of its integral points as

$$|\overline{OM} \cap \mathbb{Z}^2| = \left| \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} \in \mathbb{Z}^2 \mid 0 \leq x \leq m \right\} \right| = m + 1$$

analogously, $\overline{ON}$ has $n + 1$ integral points.

For $\overline{MN}$ we can apply Theorem 2.7.1 to get that it has $\gcd(m, n)$ integral points on it.

If we now just add up what we got for $\overline{OM}$, $\overline{ON}$ and $\overline{MN}$ we would count each vertex twice, once for each edge containing it. So if we account for that, we get:

$$\begin{aligned} |\partial T_{mn} \cap \mathbb{Z}^d| &= |(\overline{OM} + \overline{ON} + \overline{MN}) \cap \mathbb{Z}^d| \\ &= |\overline{OM} \cap \mathbb{Z}^d| + |\overline{ON} \cap \mathbb{Z}^d| + |\overline{MN} \cap \mathbb{Z}^d| - |\{O, M, N\}| \\ &= (m + 1) + (n + 1) + (\gcd(m, n) + 1) - 3 \\ &= m + n + \gcd(m, n) \end{aligned}$$

which means that

$$q(m, n) = \frac{m + n + \gcd(m, n)}{2}$$

and therefore

$$\mathbf{ehr}_{T_{mn}}(t) = 1 + \frac{m + n + \gcd(m, n)}{2}t + \frac{mn}{2}t^2.$$

5 Piecewise Ehrhart Quasi-Polynomials

In chapter 3 we described how to compute the Ehrhart polynomial of a polytope, for both integral and rational polytopes. In chapter 4 we computed Ehrhart polynomials defined by linear inequalities $Ax \leq b$ by letting some components of b vary and we introduced each component of b that we varied as a variable into the Ehrhart polynomial.

In this chapter we will describe in full generality how one can compute the *piecewise Ehrhart quasi-polynomial* $\mathbf{ehr}^A(b)$ associated to the integral matrix A of size $n \times d$.

5.1 Convention

So far, t was used to denote the scaling parameter of any Ehrhart quasi-polynomial. But Theorem 3.4.1 yields

$$\mathbf{ehr}_{P_A(b)}(t) = \mathbf{ehr}_{tP_A(b)}(1) \quad \forall t \in \mathbb{N}.$$

With some manipulations we see

$$tP_A(b) = \{t \cdot x \mid Ax \leq b\} = \{x \mid Ax \leq tb\} = P_A(t \cdot b) \quad \forall t \in \mathbb{N}.$$

Therefore

$$\mathbf{ehr}_{P_A(b)}(t) = \mathbf{ehr}_{P_A(t \cdot b)}(1) \quad \forall t \in \mathbb{N}.$$

So, when we now let every component of b vary, we no longer need a dedicated scaling variable and will therefore going forward leave out t .

5.2 Secondary Fan

In the examples shown in chapter 4, multiple polynomials were needed to fully describe the behaviour of the Ehrhart polynomials since the shape of $P_A(b)$ may change when we vary b . But we also saw Example 3.5, so in general we would also require multiple quasi-polynomials to encapsulate the Ehrhart quasi-polynomial of a polytope. This leads us to the following definition:

Definition 5.2.1. [9] *A function $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is a piecewise quasi-polynomial of degree d if and only if there exists a finite family $(U_i)_{i \leq m}$ such that $D = \cup_{i \leq m} U_i$ and a finite set of multivariate quasi-polynomials $(f_i)_{i \leq m}$ all of degree at most d , such that $f(x) = f_i(x)$ if and only if $x \in U_i$.*

The matrix A having size $n \times d$ means that in general we expect $\mathbf{ehr}^A(b)$ to be a piecewise quasi-polynomial of degree at most d in n variables $b := (b_1, \dots, b_n)$.

For the decomposition of D into $(U_i)_{i \leq m}$ to be useful, we need $P_A(b)$ and $P_A(c)$ to have the same *combinatorial structure*, whenever b and c lie in the same U_i , i.e., if $P_A(b)$ is a triangle, $P_A(c)$ must also be a triangle with the same angles, but the side lengths can differ. This is exactly the case whenever $P_A(b)$ and $P_A(c)$ have the same *normal fan* [10].

Definition 5.2.2. [6] *Given the polytope*

$$P = \{x \in \mathbb{R}^d \mid \langle u_F, x \rangle \geq -b_F\}.$$

We define the normal fan of P to be

$$\Sigma_P = \{\sigma_Q \subset \mathbb{R}^d \mid Q \preceq P\}$$

where

$$\sigma_Q = \text{Cone}(u_F \mid Q \preceq F) := \left\{ \sum_{\substack{u_F \\ Q \preceq F}} \lambda_{u_F} \cdot u_F \mid \lambda_{u_F} \geq 0 \right\}$$

are the cones of the normal fan.

We also know that for a fixed matrix A but varying b there are finitely many normal fans $\Sigma_{P_A(b)}$ [10]. Therefore, there exists a finite family $(U_i)_{i \leq m}$, which we can use to define $\text{ehr}^A(b)$.

This exact decomposition of $D \subset \mathbb{R}^n$ into the U_i can be computed and accessed via the maximal cones σ_i of the secondary fan Σ_{GKZ} of A [6].

Definition 5.2.3. [6] Let A be a matrix of size $n \times d$ and Σ_{GKZ} its secondary fan. The dimension $\dim(\sigma)$ of a cone $\sigma = \text{Cone}(h_1, \dots, h_k) \in \Sigma_{GKZ}$ is defined as the dimension of the smallest linear subspace $W = \text{Span}(h_1, \dots, h_k) \subset \mathbb{R}^n$ that contains σ . We call $\mathbb{R}^n \supset \sigma \in \Sigma_{GKZ}$ a maximal cone if

$$\dim(\sigma) := \dim(W) = n.$$

Theorem 5.2.4. [6] Let A be a matrix of size $n \times d$, Σ_{GKZ} its secondary fan and $\sigma \in \Sigma_{GKZ}$ a maximal cone. Then $P_A(b)$ and $P_A(c)$ have the same normal fan for all b and c in the interior of σ . $\square$

Definition 5.2.5. Given the secondary fan Σ_{GKZ} , we call the subset

$$\mathbb{R}^n \supset L := \bigcap_{\substack{\sigma \in \Sigma_{GKZ} \\ \sigma \text{ maximal}}} \sigma$$

the lineality space of Σ_{GKZ} .

Proposition 5.2.6. Given the secondary fan Σ_{GKZ} . $L \simeq \mathbb{R}^m$ for some $m \leq n$. $\square$

We write the basis of L as $(\lambda_1, \dots, \lambda_l) \subset \mathbb{R}^n$. For a maximal cone $\sigma = \text{Cone}(h_1, \dots, h_k) \in \Sigma_{GKZ}$, we may rewrite it to $\sigma = \text{Cone}(\pm \lambda_1, \dots, \pm \lambda_l, r_1, \dots, r_m)$, such that $2l + m = k$. We call the r_i the rays of σ and define the matrices M_σ and N_σ

$$M_\sigma := (r_1, \dots, r_m), \quad N_\sigma := (r_1, \dots, r_m, \lambda_1, \dots, \lambda_l)^{-1}$$

where M_C sends $x \in \mathbb{R}^m$ to the unique $M_\sigma \cdot x \in \sigma$ and N_C is the change of basis matrix from σ to the standard basis of $\mathbb{R}^n$.

Proposition 5.2.7. Let $A \in \mathbb{Z}^{n \times d}$ and Σ_{GKZ} its secondary fan. The basis vectors λ_i of L and the rays $r_{j,i}$ of any maximal cone $\sigma_j \in \Sigma_{GKZ}$ can be chosen to lie in $\mathbb{Z}^n$. $\square$

For a precise construction of the normal and secondary fan see §2.3. and §14.4. in [6].

5.3 Theory

As for integral and rational polytopes, we have a theorem stating the existence of the Ehrhart quasi-polynomial associated to $A \in \mathbb{Z}^{n \times d}$ on maximal cones of Σ_{GKZ} , which when computed on all maximal cones can be used to construct $\mathbf{ehr}^A(b)$.

Besides A having integer values, we also require $P_A(b)$ to be bounded for all $b \in \mathbb{Z}^n$, this is the case when $\text{Cone}(A^T) = \mathbb{R}^n$, where $\text{Cone}(A^T)$ is the cone spanned by the row vectors of A [10].

5.3.1 Computation on a Cone

For this section fix the matrix $A \in \mathbb{Z}^{n \times d}$ with $\text{Cone}(A^T) = \mathbb{R}^n$ and the maximal cone

$$C = \text{Cone}(\lambda_1, \dots, \lambda_l, r_1, \dots, r_{k-l}) = \text{Cone}(h_1, \dots, h_k) \in \Sigma_{GKZ}$$

in the secondary fan of A , on which we want to compute q_C such that

$$q_C(b) = \mathbf{ehr}^A(b) = |P_A(b) \cap \mathbb{Z}^n| \quad \forall b \in C.$$

To state the mentioned theorem we need to introduce $\hat{\text{den}}(P)$, which for a rational polytope P is the smallest positive rational number such that $\hat{\text{den}}(P)P$ is an integral polytope. And for $b \in \mathbb{R}^n$, $J \in \mathbb{N}^n$ we will write b^J and $|J|_1$ to mean:

$$b^J := \prod_{i=1}^n b_i^{J_i}, \quad |J|_1 := \sum_{i=1}^n J_i.$$

Theorem 5.3.1. [10] *Let $C = \text{Cone}(h_1, \dots, h_k) \in \Sigma_{GKZ}$ be a maximal cone. Then q_C is a quasi-polynomial in $b \in C \cap \mathbb{Q}^n$, that is,*

$$q_C(b) = \sum_{\substack{J \in \{0, \dots, d\}^n \\ |J|_1 \leq d}} c_J(A, b) b^J$$

where $c_J(A, b)$ is a piecewise-defined polynomial of total degree $n - |J|_1$ in b for all J , and $c_J(A, b) = c_J(A, b + \hat{\text{den}}(P_A(h_i)) \cdot h_i)$ for all J and i . $\square$

Note: for $C = \text{Cone}(h_1, \dots, h_k) \in \Sigma_{GKZ}$ to be a maximal cone, we need that $k \geq m$.

The theorem states the existence of q_C as a quasi-polynomial in $b \in C \cap \mathbb{Q}^n$. We make the simplification to compute q_C as a quasi-polynomial in $b \in C \cap \mathbb{Z}^n$. For this we only need to modify the $c_J(A, b)$, which are quasi-polynomials with periods $\hat{\text{den}}(P_A(h_i)) \cdot h_i \in \mathbb{Q}^n$ for all J and i , to have periods in $\mathbb{Z}^n$. To compute this integer period we use the following theorem:

Theorem 5.3.2. *Given a rational polytope P ,*

$$\hat{\text{den}}(P) = \frac{\text{per}(P)}{\text{gcd}(\text{per}(P)P)}$$

Proof. By definition, both $\hat{\text{den}}(P) \in \mathbb{Q}$ and $\text{per}(P) \in \mathbb{N}$ are positive and minimal, hence $0 < \hat{\text{den}}(P) \leq \text{per}(P)$ and we can write $\hat{\text{den}}(P)$ as a fraction p/g with $p, g \in \mathbb{N}$ coprime.

$\hat{\text{den}}(P)P = p/gP$ is an integral polytope, as is $pP = g(p/gP)$ since $g \in \mathbb{N}$. Further, we can conclude that $pP = \text{per}(P)P$, since otherwise either $\hat{\text{den}}(P)$ or $\text{per}(P)$ would not be minimal. Therefore $p = \text{per}(P)$.

Hence g is the largest integer, such that $\text{per}(P)/gP = 1/g(\text{per}(P)P)$ is still an integral polytope, which is precisely $\text{gcd}(\text{per}(P)P)$ and the claimed equality follows. $\square$

From this theorem it follows that the $c_J(A, b)$ also have periods

$$\text{per}(P_A(h_i)) \cdot h_i = \hat{\text{den}}(P_A(h_i)) \cdot \text{gcd}(\text{per}(P_A(h_i))) \cdot h_i \quad \forall J \text{ and } i.$$

Further $q_C(x + \lambda) = q_C(x)$ for all $x \in C, \lambda \in L$, which means that q_C restricted to $C \cap \mathbb{Z}^n$ is periodic on the lattice Λ generated by $(\text{per}(P_A(r_i)) \cdot r_i)_{1 \leq i \leq m}$ which is an integral lattice due to Proposition 5.2.7. From this it follows that we can interpolate q_C as a polynomial in $m \leq n$ variables and then apply a change of variables to get its representation as a quasi-polynomial over $\mathbb{R}^n$.

To have the multivariate quasi-polynomial in the form of formula (2), we need the set of off-set points for which we interpolate the multivariate polynomials, which given the fundamental parallelepiped Π of Λ is:

$$O_C := \{o_j\} = \Pi \cap \mathbb{Z}^n \simeq \mathbb{Z}^n / \Lambda. \quad (4)$$

Due to the caveat mentioned in Remark 2.2.3, we need to ensure that we can generate suitable sets of points inside C to perform the interpolation on. We can take the integral points of the d^{th} dilate of the m dimensional simplex

$$\Delta_m^d := \left\{ (x_i) \in \mathbb{Z}^m \mid \sum_{i \leq m} x_i \leq d \right\} \quad (5)$$

as a starting point, since these are exactly $\binom{d+m}{m}$.

To perform the multivariate interpolation, we also need to know the evaluation of our wanted quasi-polynomial in the interpolation points, which, given an off-set point o_j , is:

$$|P_A(M_C \times (\delta_i + o_j)) \cap \mathbb{Z}^d| \quad \forall \delta_i \in \Delta_m^d.$$

Hence, we can compute $\tilde{p}_{o_j}$ for every off-set point $o_j \in O_C$ as the multivariate interpolation polynomial of

$$\left\{ \left(\delta_i, |P_A(M_C \times (\delta_i + o_j)) \cap \mathbb{Z}^d| \right) \mid \delta_i \in \Delta_m^d, 1 \leq i \leq \binom{d+m}{m} \right\}$$

which is a polynomial in m variables. To have it as polynomial in n variables $\mathbf{x} := (x_1, \dots, x_n)$, we compute

$$p_{o_j} = \tilde{p}_{o_j} \left((N_C \times (\mathbf{x} - M_C \times o_j))_{1 \leq i \leq m} \right)$$

After computing p_{o_j} for all $o_j \in O_C$, we can piece together the wanted multivariate quasi-polynomial q_C as described in formula (2), consisting of $\pi = |\Pi \cap \mathbb{Z}^n|$ polynomials:

$$q_C(b) = \begin{cases} p_{o_1}(b) & \text{if } \bar{b} = o_1 \\ p_{o_2}(b) & \text{if } \bar{b} = o_2 \\ \vdots & \\ qp_{o_\pi}(b) & \text{if } \bar{b} = o_\pi \end{cases}$$

5.3.2 Global Computations

Our wanted piecewise quasi-polynomial $\mathbf{ehr}^A(b)$, can now be defined as

$$\mathbf{ehr}^A(b) = \begin{cases} q_{C_1}(b) & \text{if } b \in C_1 \\ q_{C_2}(b) & \text{if } b \in C_2 \\ \vdots & \\ q_{C_t}(b) & \text{if } b \in C_t \end{cases}$$

where $C_i \in \Sigma_{GKZ}$ are the maximal cones of the secondary fan of A .

By construction of the secondary fan, the maximal cones only give a decomposition of the subset $D \subset \mathbb{R}^n$ on which $P_A(b)$ is non-empty. Therefore we extend $\mathbf{ehr}^A$ to $\mathbb{Z}^n$ by setting

$$\mathbf{ehr}^A(b) = 0 \quad \forall b \notin D \cap \mathbb{Z}^n.$$

5.4 Algorithm

The algorithm to compute a piecewise Ehrhart quasi-polynomial associated to the matrix A expands upon the algorithm described in section 3.2.2, but with the notable difference, that for every maximal cone C in the secondary fan Σ_{GKZ} of A , we need to interpolate one multivariate quasi-polynomial.

Let A be an integral matrix of size $n \times d$, this means that the ambient dimension is d , which is an upper bound for the degree of the piecewise Ehrhart quasi-polynomial, and the number of variables will be n .

PiecewiseEhrhartQuasiPolynomial(A):

 if $\text{Cone}(A^T) \neq \mathbb{R}^n$:

abort

$\Sigma_{GKZ} \leftarrow \text{secondaryFan}(A)$

$L = \text{Span}(\lambda_1, \dots, \lambda_l) \leftarrow \text{linealitySpace}(\Sigma_{GKZ})$

 for $C \in \text{maxCones}(\Sigma_{GKZ})$:

$\{\pm\lambda_1, \dots, \pm\lambda_l, r_1, \dots, r_m\} \leftarrow \text{basis}(C)$

$\Lambda_C \leftarrow \text{Lattice}(r_1, \dots, r_m)$

$O_C \leftarrow \mathbb{Z}^n / \Lambda_C$

 for $o \in O_C$:

 for $p \in \Delta_m^d$:

$\text{count}_{p+o} \leftarrow |P_A(M_C \times (p+o)) \cap \mathbb{Z}^d|$

$X_o := \{(p, \text{count}_{p+o}) \mid \forall p \in \Delta_m^d\}$

```

 $p_o \leftarrow \text{interpolateMultivariatePolynomial}(X_o, m, d)$ 
 $\mathbf{x} \leftarrow (x_1, \dots, x_n) \text{ variables}$ 
 $q_o \leftarrow p_o((N_C \cdot (\mathbf{x} - M_C \cdot o))_{1 \leq i \leq m})$ 
 $q_C \leftarrow \{(o, q_o) \mid \forall o \in O_C\}$ 
return  $\{(C, q_C) \mid \forall C \in \text{maxCones}(\Sigma_{GKZ})\}$ 

```

where $\text{interpolateMultivariatePolynomial}(X_o, m, d)$ returns the interpolating multivariate polynomial of X_o in m variables of degree at most d , as described in Construction 2.2.2.

Further, having computed the Ehrhart quasi-polynomial q_C on a cone, one can check if this multivariate quasi-polynomial can be reduced to a superlattice $\Lambda'_C \supset \Lambda_C$ such that it is still periodic on Λ'_C . This would decrease the number of polynomials q_o needed to represent q_C , since Λ'_C has more points than Λ_C and therefore the quotient $\mathbb{Z}^n/\Lambda'_C$ has fewer points than $\mathbb{Z}^n/\Lambda_C$.

6 Package

To go along with this thesis and demonstrate the correctness of the shown algorithms, a Sage package has been developed, which can be found on [GitHub](#) [19]. Its main use case is to compute piecewise Ehrhart quasi-polynomials associated to matrices but it also provides class implementations of integer periodic functions and quasi-polynomials and an implementation to compute Ehrhart (quasi-)polynomials.

6.1 Contents of the Package

The package contains the following parts:

1. implementation of integer periodic functions as described in section 2.3
2. implementation of quasi-polynomials as described in section 2.4
3. an algorithm to compute the Ehrhart quasi-polynomials of polytopes as described in the sections 3.1.2 and 3.2.2
4. implementation of piecewise Ehrhart quasi-polynomials associated to matrices as described in section 5.4

6.2 Installation

The package was is designed to work with Sage version 10.3 or newer installed in a virtual environment inside a *Linux* terminal. For a guide to install sage see [18].

Open a terminal and clone the git repository using (do not type the \$)

```
$ git clone https://github.com/mwooll/EhrhartQuasiPolynomial.git
```

inside the new directory named “/EhrhartQuasiPolynomial” run

```
$ sage -pip install .
```

this installs the package globally.

6.3 Getting Started

The package can be used to compute the Ehrhart quasi-polynomials of rational polytopes as in section 3.5 and evaluating them afterwards as shown in Listing 1. The returned objects from `compute_ehrhart_quasi_polynomial` are fully fleshed out instances of quasi-polynomials which know what their period and degree is, and can be used for computations involving other quasi-polynomials and constants, see Listing 2.

```
1 sage: from ehrhart_quasi_polynomial import  
    compute_ehrhart_quasi_polynomial  
2 sage: rational_triangle = [(0, 0), (1, 0), (0, 0.5)]  
3 sage: f = compute_ehrhart_quasi_polynomial(rational_triangle); print(f)  
4 [1, 3/4] + t + 1/4*t^2  
5 sage: [f(n) for n in range(1, 7)]
```

```
6 [2, 4, 6, 9, 12, 16]
```

Listing 1: Recreation of Example 3.5.

```
1 sage: f.period(), f.degree()
2 (2, 2)
3 sage: from ehrhart_quasi_polynomial import QuasiPolynomialRing
4 sage: QPR = QuasiPolynomialRing(QQ)
5 sage: g = QPR([[0, 1, 2], [0, 1], [1]]); print(g)
6 [0, 1, 2] + [0, 1]*t + t^2
7 sage: print(2*f + g - 2)
8 [0, 1/2, 2, -1/2, 1, 3/2] + [2, 3]*t + 3/2*t^2
9 sage: h = f*g; print(h)
10 [0, 3/4, 2, 0, 1, 3/2] + [0, 7/4, 2, 3/4, 1, 11/4]*t + [1, 2, 3/2, 7/4,
    5/4, 9/4]*t^2 + [1, 5/4]*t^3 + 1/4*t^4
11 sage: h.period(), h.degree()
12 (6, 4)
```

Listing 2: Continuation of Listing 1

But the main functionality of the package is to provide an implementation of the algorithm to compute piecewise Ehrhart quasi-polynomials, section 5.4 reflects this implementation.

```
1 sage: from ehrhart_quasi_polynomial import
    PiecewiseEhrhartQuasiPolynomial
2 sage: A = Matrix([[ -1, 0], [0, -1], [1, 1]])
3 sage: p = PiecewiseEhrhartQuasiPolynomial(A)
4 sage: print(p)
5 On Cone((1, 1, 1), (1, 0, -1), (0, 1, -1), (-1, 0, 1), (0, -1, 1))
6 invariant on Lattice((1, 0, 0), (1, 1, 1), (1, 0, -1), (0, 1, -1)):
7 1/2*x0^2 + x0*x1 + 1/2*x1^2 + x0*x2 + x1*x2 + 1/2*x2^2 + 3/2*x0 +
  3/2*x1 + 3/2*x2 + 1 if [b] = (0, 0, 0)
8 sage: [p((0, 0, b)) for b in [1, 2, 5, 10]]
9 [3, 6, 21, 66]
10 sage: def ehr_Tb(b, t): return 1/2*b^2*t^2 + 3/2*b*t + 1
11 sage: [ehr_Tb(b, 1) for b in [1, 2, 5, 10]]
12 [3, 6, 21, 66]
```

Listing 3: Continuation of Example 4.1

As shown in Listing 3, we receive $\mathbf{ehr}_{T_b}(1)$ from $\mathbf{p}$, when we set $\mathbf{x}_1$ and $\mathbf{x}_2$ to 0 and rename $\mathbf{x}_3$ to $\mathbf{b}$, which was the setting in section 4.1.

6.4 Example Computations

As we have seen in section 4.2, $\mathbf{ehr}_{Q_{mn}}(t)$ is a piecewise polynomial in m, n and t , consisting of 2 non-zero polynomials. In fact, even if consider

$$P = \left\{ \binom{x}{y} \mid \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} \cdot \binom{x}{y} \leq \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} \right\},$$

i.e., rename m and n to x_2 and x_3 , respectively, and introduce 2 new variables x_0 and x_1 , 2 non-zero polynomials still suffice to capture the full behaviour of the piecewise Ehrhart quasi-polynomial of P :

```

1 sage: from ehrhart_quasi_polynomial import
    PiecewiseEhrhartQuasiPolynomial
2 sage: A = Matrix([[ -1, 0], [0, -1], [1, 1], [0, 1]])
3 sage: p = PiecewiseEhrhartQuasiPolynomial(A); print(p)
4 On Cone((-1, 2, -1, 3), (1, 3, 1, 2), (1, 0, -1, 0), (0, 1, -1, -1),
    (-1, 0, 1, 0), (0, -1, 1, 1))
5 invariant on Lattice((3, 0, -1, 0), (0, 1, -1, 2), (-1, 2, -1, 3), (1,
    3, 1, 2), (1, 0, -1, 0), (0, 1, -1, -1)):
6 1/2*x0^2 + x0*x1 + 1/2*x1^2 + x0*x2 + x1*x2 + 1/2*x2^2 + 3/2*x0 +
    3/2*x1 + 3/2*x2 + 1 if [b] = (0, 0, 0, 0)
7
8 On Cone((1, 3, 1, 2), (2, 1, 2, -1), (1, 0, -1, 0), (0, 1, -1, -1), (-1,
    0, 1, 0), (0, -1, 1, 1))
9 invariant on Lattice((1, 1, 0, 1), (0, 3, 1, 3), (1, 3, 1, 2), (2, 1, 2,
    -1), (1, 0, -1, 0), (0, 1, -1, -1)):
10 x0*x1 + 1/2*x1^2 + x1*x2 + x0*x3 + x2*x3 - 1/2*x3^2 + x0 + 3/2*x1 +
    x2 + 1/2*x3 + 1 if [b] = (0, 0, 0, 0)

```

Listing 4: Continuation of Example 4.2

The connections between this output and Figure 7 are the following:

- The red area is contained in the first cone mentioned and notably the polynomial on this cone is independent of x_3 , i.e., n .
- The green area is contained in the second cone and as in section 4.2 this is a polynomial in all possible variables.
- The blue area is suppressed in the output since it makes the convention that a piecewise (quasi-)polynomial is zero outside the stated cones.
- Setting x_0 and x_1 to 0 and renaming x_2 to m and x_3 to n , yields the polynomials as shown in Figure 7.

```

1 sage: from ehrhart_quasi_polynomial import
    PiecewiseEhrhartQuasiPolynomial
2 sage: B = Matrix([[ -1, 0], [0, -1], [1, 2], [0, 1]])
3 sage: q = PiecewiseEhrhartQuasiPolynomial(B); print(q)
4 On Cone((-1, 1, -1, 3), (1, 3, 1, 1), (1, 0, -1, 0), (0, 1, -2, -1),
    (-1, 0, 1, 0), (0, -1, 2, 1))
5 invariant on Lattice((0, 0, 2, 0), (0, 0, 0, 1), (-1, 1, -1, 3), (1, 3,
    1, 1), (1, 0, -1, 0), (0, 1, -2, -1)):
6 1/4*x0^2 + x0*x1 + x1^2 + 1/2*x0*x2 + x1*x2 + 1/4*x2^2 + x0 + 2*x1 +
    x2 + 1 if [b] = (0, 0, 0, 0)
7 1/4*x0^2 + x0*x1 + x1^2 + 1/2*x0*x2 + x1*x2 + 1/4*x2^2 + x0 + 2*x1 +
    x2 + 3/4 if [b] = (0, 0, 1, -1)
8
9 On Cone((1, 1, 1, -1), (1, 3, 1, 1), (1, 0, -1, 0), (0, 1, -2, -1), (-1,
    0, 1, 0), (0, -1, 2, 1))
10 invariant on Lattice((0, 1, 1, 0), (0, 0, 0, 1), (1, 1, 1, -1), (1, 3,
    1, 1), (1, 0, -1, 0), (0, 1, -2, -1)):
11 x0*x1 + x1^2 + x1*x2 + x0*x3 + x2*x3 - x3^2 + x0 + 2*x1 + x2 + 1
    if [b] = (0, 0, 0, 0)

```

Listing 5: Example Piecewise Ehrhart Quasi-Polynomial

In the general case, one would expect the result to be a true piecewise quasi-polynomial in b and as Listing 5 shows, the periods on the different cones can be different.

6.5 Correctness

```

1 def test_PEQP(A, peqp, min_val, max_val, quiet=True):
2     """
3     Return True whether ``peqp`` evaluated at `b` in the n-cube ``[
4     min_val, max_val]^n`` is equal to the number of integral points of `
5     P_A(b)` = ``create_polytope_from_matrix(A, b)``.
6     """
7     if A.nrows() != peqp._amb_dim:
8         raise ValueError(f"A has dimensions {A.nrows()}x{A.ncols()} but "
9                           f"peqp is a polynomial in {peqp._amb_dim}
10                          variables.")
11
12     ehr = lambda b: len(create_polytope_from_matrix(A, b).
13                          integral_points())
14
15     for b in combinations_with_replacement(range(min_val, max_val+1),
16                                             peqp._amb_dim):
17         e = ehr(b)
18         p = peqp(b)
19         if not quiet:
20             print(f"{b}: {e} == {p} is {e == p}")
21         if e != p:
22             print(f"Different evaluation found for b = {b}: "
23                   f"f"ehr(b) = {e} != {p} = peqp(b)")
24             return False
25
26     return True

```

Listing 6: Function to test Correctness

To demonstrate the correctness of the shown piecewise Ehrhart quasi-polynomials in section 6.4, we use the function `test_PEQP` shown in Listing 6, which is also provided in the package. In `test_PEQP`, `ehr(b)` computes the number of integral points of $P_A(b)$, i.e., $|P_A(b) \cap \mathbb{Z}^d|$ and this value is compared with the evaluation `peqp(b)` of the provided `PiecewiseEhrhartQuasiPolynomial` for all $b \in \text{value_range}^n$, where n is the number of rows of A .

```

1 sage: from ehrhart_quasi_polynomial import
2       PiecewiseEhrhartQuasiPolynomial, test_PEQP
3 sage: A = Matrix([-1, 0], [0, -1], [1, 1], [0, 1])
4 sage: p = PiecewiseEhrhartQuasiPolynomial(A)
5 sage: test_PEQP(A, p, 1, 3, False)
6 (1, 1, 1, 1): 9 == 9 is True
7 (1, 1, 1, 2): 10 == 10 is True
8 (1, 1, 1, 3): 10 == 10 is True
9 (1, 1, 2, 2): 14 == 14 is True
10 (1, 1, 2, 3): 15 == 15 is True
11 (1, 1, 3, 3): 20 == 20 is True
12 (1, 2, 2, 2): 20 == 20 is True
13 (1, 2, 2, 3): 21 == 21 is True
14 (1, 2, 3, 3): 27 == 27 is True
15 (1, 3, 3, 3): 35 == 35 is True
16 (2, 2, 2, 2): 25 == 25 is True
17 (2, 2, 2, 3): 27 == 27 is True
18 (2, 2, 3, 3): 33 == 33 is True
19 (2, 3, 3, 3): 42 == 42 is True
20 (3, 3, 3, 3): 49 == 49 is True

```

```

21 sage: test_PEQP(A, p, -10, 10, True)
22 True

```

Listing 7: Correctness of p from Listing 4

```

1 sage: from ehrhart_quasi_polynomial import
    PiecewiseEhrhartQuasiPolynomial, test_PEQP
2 sage: B = Matrix([[ -1, 0], [0, -1], [1, 2], [0, 1]])
3 sage: p = PiecewiseEhrhartQuasiPolynomial(B)
4 sage: test_PEQP(B, q, -10, 10, True)
5 True

```

Listing 8: Correctness of q from Listing 5

Listing 7 and Listing 8 verify that the number of integral points of $P_A(b)$ and $P_B(b)$, are all equal to the evaluation of p for all $b \in [-10, 10]^3$ and q for all $b \in [-10, 10]^4$, respectively.

```

1 sage: from ehrhart_quasi_polynomial import
    PiecewiseEhrhartQuasiPolynomial, test_PEQP
2 C = Matrix([[ -1, 0], [0, -1], [2, 0], [0, 3]])
3 sage: r = PiecewiseEhrhartQuasiPolynomial(C); print(r)
4 On Cone((0, 3, 0, 1), (2, 0, 1, 0), (1, 0, -2, 0), (0, 1, 0, -3), (-1,
    0, 2, 0), (0, -1, 0, 3))
5 invariant on Lattice((2, 0, 2, 0), (0, 6, 0, 3), (0, 9, 0, 3), (4, 0, 2,
    0), (1, 0, -2, 0), (0, 1, 0, -3)):
6   x0*x1 + 1/2*x1*x2 + 1/3*x0*x3 + 1/6*x2*x3 + x0 + x1 + 1/2*x2 + 1/3*
    x3 + 1 if [b] = (0, 0, 0, 0)
7   x0*x1 + 1/2*x1*x2 + 1/3*x0*x3 + 1/6*x2*x3 + 1/3*x0 + x1 + 1/6*x2 +
    1/3*x3 + 1/3 if [b] = (0, -2, 0, -1)
8   x0*x1 + 1/2*x1*x2 + 1/3*x0*x3 + 1/6*x2*x3 + 2/3*x0 + x1 + 1/3*x2 +
    1/3*x3 + 2/3 if [b] = (0, -4, 0, -2)
9   x0*x1 + 1/2*x1*x2 + 1/3*x0*x3 + 1/6*x2*x3 + x0 + 1/2*x1 + 1/2*x2 +
    1/6*x3 + 1/2 if [b] = (-1, 0, -1, 0)
10  x0*x1 + 1/2*x1*x2 + 1/3*x0*x3 + 1/6*x2*x3 + 1/3*x0 + 1/2*x1 + 1/6*x2
    + 1/6*x3 + 1/6 if [b] = (-1, -2, -1, -1)
11  x0*x1 + 1/2*x1*x2 + 1/3*x0*x3 + 1/6*x2*x3 + 2/3*x0 + 1/2*x1 + 1/3*x2
    + 1/6*x3 + 1/3 if [b] = (-1, -4, -1, -2)
12 sage: test_PEQP(C, r, -10, 10, True)
13 True

```

Listing 9: Interesting Piecewise Quasi-Polynomial

Listing 9 shows the correctness $\mathbf{ehr}^C(b)$ for the matrix $C = \begin{pmatrix} -1 & 0 \\ 0 & -1 \\ 2 & 0 \\ 0 & 3 \end{pmatrix}$ for all $b \in$

$[-10, 10]^4$ and the fact that the quasi-polynomial can have multiple differing monomials which may vary independently for each point.

6.6 Discussion of Conditions

The algorithm to compute piecewise Ehrhart quasi-polynomials associated to matrices is only defined for integral matrices A such that $\text{Cone}(A^T) = \mathbb{R}^n$ and computes piecewise quasi-polynomials which are only guaranteed to have meaningful evaluations, i.e. integers, for integer vector arguments.

6.6.1 $\text{Cone}(A^T) = \mathbb{R}^n$

Dropping the condition $\text{Cone}(A^T) = \mathbb{R}^n$ would imply that $P_A(b)$ could be unbounded for some b , i.e. $|P_A(b) \cap \mathbb{Z}^d| = \infty$, which makes it impossible to interpolate a polynomial in a meaningful way. However, one could still compute the piecewise Ehrhart quasi-polynomial on the maximal cones C on which $P_A(b)$ is bounded for all $b \in C$ and return ∞ if b belongs to a cone on which $P_A(b)$ is unbounded. Though this seems like a rarely helpful compromise and was hence left out of the implementation.

6.6.2 $A \in \mathbb{Z}^{n \times d}$

Allowing A to be a matrix with rational entries brings up similar problems as wanting the quasi-polynomials to return integers for rational argument vectors, even though Theorem 5.3.1 provides the lattice Λ on which the quasi-polynomial would be periodic, since in order to define the quasi-polynomial we need to interpolate a polynomial for every representative of the quotient $\mathbb{Z}^n/\Lambda$.

If we just replace $\mathbb{Z}^n$ with $\mathbb{Q}^n$ the quotient would always be infinite, since Λ is discrete. Computing an infinite amount of polynomials is obviously not possible in a finite amount of time. And no matter which superlattice Λ' of $\mathbb{Z}^n$ and Λ we take, such that the quotient Λ'/Λ is finite, there will always be only finitely many polynomials which cannot guarantee that all rational evaluations yield integers.

In [10] the authors give examples of functions defined on $\mathbb{Q}^4$ which apparently always evaluate to integers. But since they also make use of the fractional part of the input, which is not possible to mimic using quasi-polynomials, these are not quasi-polynomials and hence it is not clear how their example could be generalized to work on arbitrary matrices and be used as the foundation of a new algorithm.

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